

BIO 1380 Cells, Energy and Heredity

Practical handbook

School of Life Sciences, Tarnhollow University

Autumn 2026. Second issue, Friday 4 September 2026. Dr. Maelis Underhaye, with the teaching laboratory team.

Keep this handbook for the whole term. It says how the practical classes run, what each of the four practicals asks of you, and how reports are written, handed in and marked. It does not repeat the lectures. Where a practical leans on a lecture, its section says so.

1. How the practicals run

There are four practicals, P1 to P4. Each is a single session of two and a half hours in the Wetherden Teaching Laboratory, room LS 0.19, on the ground floor of the Life Sciences building. You attend with your group and you work at a bench of four people.

Group	Day	Session
A	Monday	09:10 to 11:40
B	Monday	14:10 to 16:40
C	Wednesday	09:10 to 11:40
D	Wednesday	14:10 to 16:40

Group lists are on the course site. You may not swap groups by private arrangement.

Practical	Week beginning	Subject	You hand in	Due, all groups
P1	Monday 14 September	Catalase and temperature	Short report	Friday 25 September
P2	Monday 28 September	Yeast fermentation on different sugars	Full report	Monday 12 October
P3	Monday 12 October	Scoring marsh beetle crosses	Full report	Friday 23 October
P4	Monday 2 November	Osmosis in potato tissue	Full report	Friday 13 November

The short report belongs to P1 only: one side of A4 and at most 350 words. It is there so that you get comments on your writing before the full reports begin. Full reports are described in section 9.

The people you will meet:

- Ottoline Varkess, senior demonstrator, runs every session and is the first person to ask about anything in this handbook. Room LS 0.21.
- Idris Penhallick, laboratory manager, looks after the room, the equipment, the lockers and lost property. Room LS 0.23.
- The demonstrators are postgraduate students, one to every three benches. They help at the bench and they mark the reports.

2. Before each session

- Read the section for the practical and complete the risk checklist on the course site by 18:00 on the day before your session. No checklist, no bench.
- Bring your lab coat, your bound notebook, a pen and a calculator. Phones stay in your bag, and bags go in the lockers by the door; the laboratory manager issues a locker key against your student card.
- The briefing begins at ten past the hour. If you arrive more than ten minutes after it has begun you will be asked to book the catch-up session instead.
- If you miss a session, email the senior demonstrator within 2 working days of the session you missed. A catch-up session runs on the Friday of the same week, 14:10 to 16:40, with 12 places. A missed session with no catch-up means no data, and a report cannot be written from somebody else's results.

3. Safety

The teaching laboratory is a low-hazard room, but it is still a laboratory.

- Lab coats are worn and fastened at all times. Eye protection is worn in P1, where you handle hydrogen peroxide.
- No food, no drink, no chewing gum. Water bottles stay in the corridor racks.
- Long hair is tied back. Closed shoes only.
- Broken glass goes in the blue-lidded bin, never in the ordinary waste. Tell the laboratory manager at once; nobody is in trouble for an honest breakage.
- The beetles in P3 are live animals. They are handled in their trays with the soft brush provided, and not one of them leaves the room.
- Wash your hands before you leave, every time.

Three hazards are worth knowing by name. Hydrogen peroxide at the strength we use (3% by volume) is an irritant: rinse splashes with plenty of water. The water baths at 55 and 75 degrees C in P1 will scald. Cork borers and scalpels in P1 and P4 are sharp, and are counted out and counted back in.

4. The laboratory notebook

You keep one bound notebook for all four practicals. Loose sheets are not accepted.

- Date every entry and write in ink.
- Record raw readings straight into the notebook as you take them, never on scrap paper to be copied up later.
- Cross out mistakes with a single line so that the original can still be read.
- Before you leave, a demonstrator initials your results page. A report whose data cannot be traced to an initialled page is sent back without a mark.

The notebook itself is not handed in.

5. Practical 1: catalase and temperature

Leans on: lecture 4 (enzymes).

Catalase breaks hydrogen peroxide down into water and oxygen. You measure how fast it does so at five temperatures and look for the temperature at which the rate is greatest.

Method in brief:

1. Cut potato discs 5 mm thick with the number 3 cork borer. Each tube gets six discs.
2. Stand the tubes in the water baths at 5, 20, 37, 55 and 75 degrees C. Leave each tube for 8 minutes so that the discs reach the temperature of the bath.
3. Add 5 mL of hydrogen peroxide solution, fit the delivery tube, and collect the oxygen over water in an inverted measuring cylinder.
4. Read the volume of oxygen after 90 seconds.
5. Run three replicates at each temperature, with fresh discs every time.

What to expect: the rate climbs from the coldest bath to the bath at 37 degrees C and then collapses. Be ready to explain why the fall above 37 degrees C is not simply the rise run backwards.

What goes in the short report: a results table with all three replicates, a graph of mean volume against temperature with the range drawn as bars, and one paragraph of interpretation.

6. Practical 2: yeast fermentation on different sugars

Leans on: the lecture on cellular respiration, and the flask demonstration you saw in it.

Baker's yeast is given five different things to live on, with no air, and you measure how much gas each tube gives off.

Method in brief:

1. Label five fermentation tubes: glucose, sucrose, maltose, lactose, and a control with water in place of the sugar solution.
2. To each tube add 10 mL of yeast suspension and 10 mL of sugar solution. All four sugar solutions are made up at 50 g per litre.
3. Fill the tube so that no air is trapped, invert it in its beaker, and stand the beaker in the water bath at 32 degrees C.
4. Read the volume of gas in the closed arm at 10, 25 and 40 minutes, timed from the moment the tube goes into the bath.
5. Enter your bench's readings on the class sheet before you leave. The pooled class results are put on the course site as a spreadsheet file once the last group has met.

What to expect: the tubes do not behave alike, and one sugar will hardly differ from the water control. Your discussion should say what the yeast would need in order to use each sugar. Maltose usually lags and then catches up; say why that might be.

Use the class means in your report as well as your own bench's figures, and say where your bench differs from the class. Do not try to convert a volume of gas into a number of ATP. This practical does not measure that.

7. Practical 3: scoring marsh beetle crosses

Leans on: the lecture on Mendelian genetics. Use the lecture's notation, and write ratios the way the lecture writes them.

The crosses were set up by the technicians in the teaching laboratory's insectary five weeks before your session, and the parents were taken out once the eggs had been laid. Each bench receives two trays of adult offspring.

- Tray 1 comes from a one-gene cross: both parents were heterozygous for wing-case colour.

- Tray 2 comes from a two-gene cross: both parents were heterozygous for wing-case colour and for spots.

Scoring:

1. Chill each tray on the cold plate for seven minutes so that the beetles move slowly.
2. Score every beetle in the tray, once. Move each scored beetle to the pot marked SCORED so that it cannot be counted twice.
3. Colour is scored on the wing cases only, in daylight or under the white lamp. The bench lamps with yellow bulbs make grey look russet.
4. A beetle counts as spotted if it has one clear spot or more on either wing case. Smudges at the wing edge are not spots.
5. A tray with fewer than 90 offspring is too small to analyse by itself. Ask the senior demonstrator for a second tray from the same cross and add the two together.

Analysis: for each tray, work out the numbers that the lecture's ratio predicts for your total, and compare your counts with them using chi-square. The degrees of freedom are the number of phenotype classes less one. Appendix A gives the critical values. We work at the 0.05 level throughout this course. Round chi-square to hundredths, and state in words whether your counts depart from the prediction.

Do not pool your trays with another bench's in your own report. The class sheet does the pooling.

8. Practical 4: osmosis in potato tissue

Leans on: lecture 5 (membranes and transport).

You find the sucrose concentration at which potato tissue neither gains water nor loses it.

Method in brief:

1. Cut cylinders with the number 5 cork borer and trim each one to 40 mm. Each bench needs 18 cylinders: three for each of six solutions.
2. Blot each cylinder and weigh it to 0.01 g.
3. Put three cylinders into each sucrose solution: 0.00, 0.15, 0.25, 0.35, 0.55 and 0.80 mol per litre.
4. Leave them for 55 minutes at room temperature. Do not cut the time short to catch a bus: the cylinders in the strongest solutions are still changing well into the session.
5. Blot each cylinder exactly as before and weigh it again.
6. Work out the percentage change in mass for every cylinder, and the mean for each solution.

Plot the mean percentage change against concentration and read off where your line crosses zero. Say how sure you are of that reading, and what limits it.

9. Reports

What a full report contains

Title; aim; method; results; analysis; discussion; references. The method section is short: say that you followed this handbook, and describe only what you did differently. Tables and graphs carry numbered legends.

A full report is at most 1,400 words, not counting tables, figure legends and the reference list. Print the word count on the first page. A report that runs over is read as far as the limit and marked on what has been read.

Handing in

Reports go in as a single PDF through the course site by 09:00 on the due date shown in section 1. The due date is the same for all four groups, whichever day you were in the laboratory. Name the file with your student number and the practical, for example 4471902_P3.pdf.

Demonstrators cannot change a deadline. If you cannot meet one, apply to the Life Sciences teaching office, room LS 1.02, before the deadline has passed.

Marking and feedback

Reports are marked by the demonstrators and checked by the senior demonstrator. Each report comes back with a band (Excellent, Good, Sound, Weak or Incomplete) and written comments under five headings: data, presentation, analysis, interpretation and writing.

Marked reports are returned through the course site, as a rule within two weeks of the due date. If you think a band is wrong, raise it with the senior demonstrator within 5 working days of the report being returned, and bring the report and your notebook with you. Demonstrators do not re-mark at the bench.

Working together

You collect data as a bench and you write alone. The four of you will have the same raw numbers. The tables, the graphs, the analysis and the words must be your own.

Faults that pulled reports down last term

- A graph with no units on its axes, or with the points joined dot to dot where a fitted line was wanted.
- A discussion that says the results over again in sentences and stops there.
- Chi-square given with no degrees of freedom beside it, so that the reader cannot check the conclusion.
- "Human error" offered as an explanation. Say which step, and which way it would have pushed the numbers.
- Class figures quoted with no mention of the bench's own.

Appendix A. Critical values of chi-square at the 0.05 level

Degrees of freedom	Critical value
1	3.84
2	5.99
3	7.81
4	9.49
5	11.07

If your chi-square is greater than the critical value, your counts differ from the prediction by more than chance readily explains.

Appendix B. Equipment and disposal

- Balances: read to 0.01 g. Tare with the weighing boat already on the pan.
- Water baths: set before you arrive. Do not touch the dials; if a bath reads wrong, tell a demonstrator.
- Measuring cylinders and fermentation tubes: rinse and leave upside down on the rack by the sink.

- Yeast tubes from P2: tip into the yellow-lidded bin, then rinse.
- Potato waste from P1 and P4: compost caddy, not the sink.
- Leftover peroxide: down the sink with the tap running.
- Beetles from P3: every beetle, scored or not, goes back into its own tray. The technicians collect the trays at the end of the session.

Appendix C. Who to go to

Matter	Go to
Anything about the method on the day	The demonstrator at your bench
A missed session, a change of group	Senior demonstrator
A band you think is wrong	Senior demonstrator
Breakages, locker keys, lost property	Laboratory manager
A deadline you cannot meet	Life Sciences teaching office
Access needs in the laboratory	Senior demonstrator, before the first session